Sky High Design Report

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Top-Level Design Description

Sky High is a 3D physics-based stacking arcade game where the player must create a stack of shapes as high as possible. The goal is to create the highest stack possible with many different shaped. The player is rewarded when they choose unstable shapes over stable ones. Variety is also rewarded. The player must find a balance between maximizing bonus points and building a sturdy tower. The game ends when the tower falls over.

Design Pillars

The below design pillars identify the types of fun which are key to the user experience

Application of an Ability	**Thrill of Danger**	**Intellectual Problem Solving**
The player can choose where to drop the shape so they must precisely drag and drop shape to land neatly with the margin for error shrinking as the stack grows taller.	Every placement carries a risk. The thin pieces are unstable and placing one makes it more likely the whole stack will topple.	The player must decide what shape to place and when. The player is always thinking of risk vs reward.

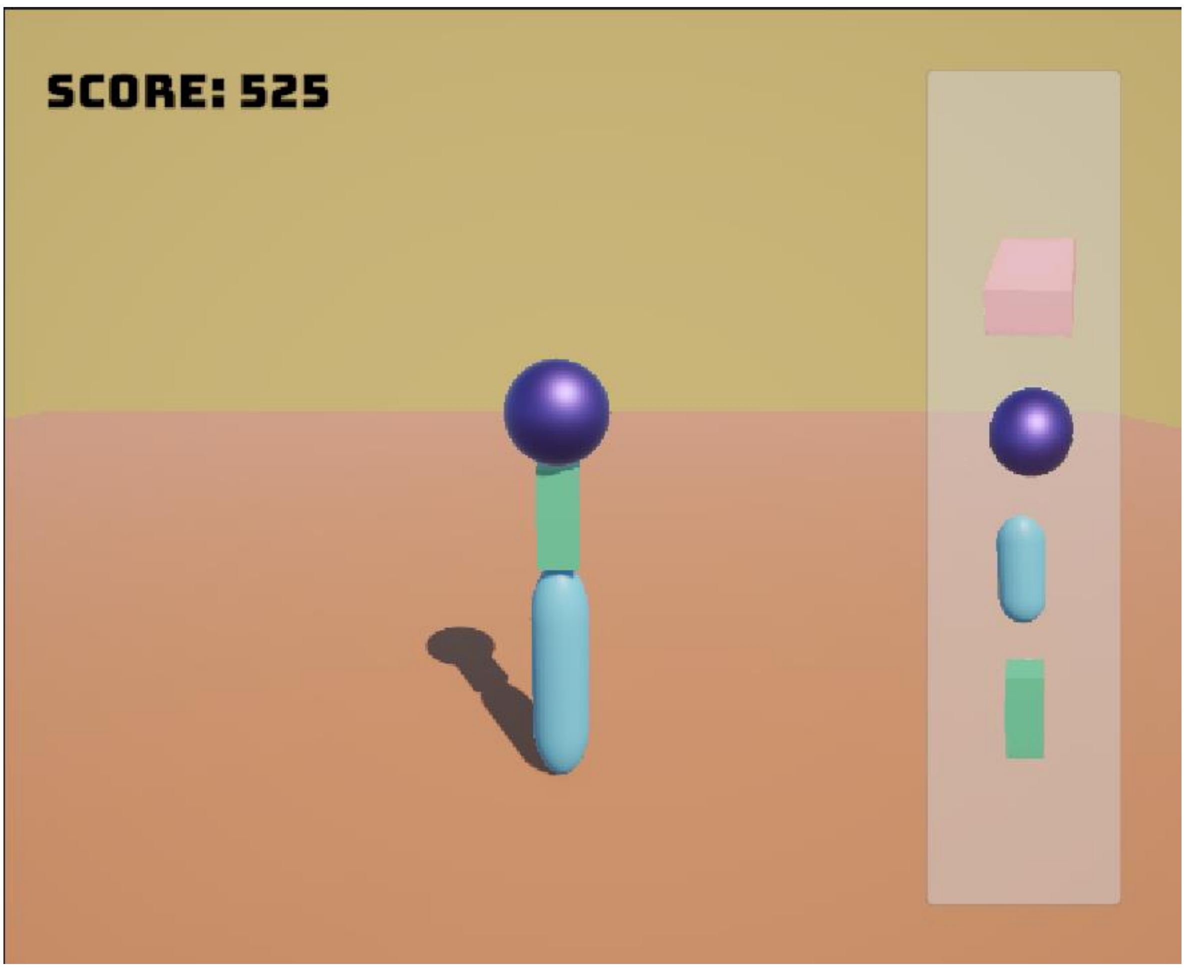
Audience & Marketing

Sky High is designed to be an offline single-player game playable in short sessions on desktop. It is a casual physics-based stacking game aimed at players who enjoy quick replayable skill challenges like Jenga.

The game is built as an original portfolio project to demonstrate systems design, risk-reward scoring and physics-driven gameplay.

Core Gameplay

The player drags shapes from a tray on the side of the screen onto the ground. By doing this they will build up a tower up piece by piece. Unstable shapes are worth more points than stable shapes. The player also earns a bonus the first time they use each new shape. The player will continue to stack shapes until the stack falls over. The game ends when the stack collapses. The game over screen displays the player's final score.



Controls

Mouse Input: Click and drag to choose and move shapes.

Gameplay Balance & Pacing

The difficulty curve emerges when the stack gets bigger as shapes become harder to balance on top. Each piece added to the stack raises the platform's center of mass so it's easier to stack in the beginning and harder later in the game.

Each shape has a different width, and different points awarded. Unstable shapes are worth significantly more points than stable shapes.

Placing a thin item low in the stack doubles its score if it stays balanced. This rewards the player for taking a real risk rather than always choosing to the safest and largest shape. The player also earns a bonus for each time they use each new shape which encourages variety rather than repeatedly placing one large stable shape.

Tone & Aesthetics

This is my first 3d game. So, I didn't set out for this to have a theme or certain aesthetic. But early on it was mostly grey and I did want to change this.

Instead of using the default skybox in Unity I coloured the floor to be brown and the background to be yellow so it's like the game takes place in a room. I added colours to all the shapes also, so they were not just grey. I did this by making a material in Unity for each new colour I wanted.

Narrative

This is an offline singleplayer game without any explicit narrative. I do not intend to have dialog or characters speaking.

Business Model

Sky High is a free-to-download Unity game with no monetisation model. It will be made freely available to all users with no ads and no premium tiers.

The game is being developed as a portfolio project to demonstrate skills in game design and development. Its primary value is in the player experience and the learning gained through building it, rather than revenue generation.